

CURRICULUM VITAE

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Education

University of Waterloo

Incoming Ph.D. Student in Computer Science

Advisor: Prof. Khuzaima Daudjee and Asst. Prof. Hong Zhang

Sept 2026 – Expected 2030

Waterloo, Canada

University of Science and Technology of China

Master of Engineering in Computer Science

Sept 2022 – Jul 2025

Hefei, China

University of Science and Technology of China

Bachelor of Science in Physics

Sept 2018 – Jul 2022

Hefei, China

Research Interests

My research interests broadly lie in Efficient Machine Learning Systems, with a particular emphasis on developing resource-aware algorithms and system architectures to enable efficient LLM serving. I have explored diverse research topics, ranging from video LLM, data valuation, and edge caching decisions, aiming to improve resource utilization efficiency and service quality in distributed systems.

Research Experience

Research Intern, advised by Asst. Prof. Banghua Zhu

Department of Electrical Computer Engineering, University of Washington

Nov 2025– Present

Seattle, America

- **Optimize the inference efficiency of diffusion language models**

- Introduce an adaptive sparsity budget allocation mechanism to minimize the computational cost of diffusion language models.
- Submitted to NIPS 2026 as the co-first author

Graduate Researcher, advised by Prof. Mingjun Xiao, Asst. Prof. Yin Xu

Department of Computer Science, USTC

Sept 2022– Present

Hefei, China

- **Collaborative VideoLLM framework for real-time video understanding on edge device**

- Proposed ECVideoLLM, a new edge-cloud collaborative serving paradigm for VideoLLMs, to enable real-time video understanding on edge devices. Designed a cloud-side query-agnostic memory construction mechanism to compactly capture long-video contexts without continuously transmitting raw video frames from edge devices.
- Led the project; manuscript in preparation for submission to IEEE INFOCOM 2027.

- **Large-scale mobile edge caching problem considering economic interplay**

- Introduced a large-scale edge caching game model jointly considering content caching, trading, and sharing in the content delivery network. Devised a mean field algorithm to approximate the collective impact of edge servers in order to reduce the computational complexity of obtaining the game equilibrium.
- One paper accepted at ICDE 2024, with me as the student first author.

- **Data-driven federated learning using drones**

- Proposed a data-driven federated learning (FL) paradigm with the abstraction of the independent data layer. Constructed data quality metrics and developed an algorithm based on Bayesian inference and the bandit model to achieve optimal scheduling on distributed unknown data. The algorithm achieved better performance on theoretical and real-world datasets.
- First author paper accepted by ICDE 2025.

Publications

* Equal contribution; † denotes project lead.

- **Step-dLLM: Adaptive Step-aware Sparse Attention for Efficient Diffusion LLM Inference**

Zhichen Zeng*, Xichong Zhang*, Junpan Wu, ..., Banghua Zhu

Under Review

- **ECVideoLLM: Communication-Efficient Online VideoLLMs via Memory-First Edge–Cloud Collaboration**

Yihe Weng*, Xichong Zhang*†, Jiabin Ou, Luyao Gao, Yin Xu, Mingjun Xiao

Working

- **Online Federated Learning on Distributed Unknown Data Using UAVs**
Xichong Zhang, Haotian Xu, Yin Xu, Mingjun Xiao, Jie Wu, and Jinrui Zhou
International Conference on Data Engineering (ICDE), 2025
- **Joint Mobile Edge Caching and Pricing: A Mean-Field Game Approach**
Yin Xu (advisor), Xichong Zhang, Mingjun Xiao, Jie Wu, An Liu, and Sheng Zhang
International Conference on Data Engineering (ICDE), 2024
- **Collaborative Federated Learning on Distributed Data Using Multiple UAVs**
Xichong Zhang, Jinrui Zhou, Yin Xu, Mingjun Xiao, Jie Wu
Submitted to IEEE/ACM Transactions on Networking
- **AoI-Aware Mobile Crowdsensing in Non-stationary and Constrained Restless Event Environments**
Xichong Zhang, Jinrui Zhou, Haotian Xu, Yin Xu, Mingjun Xiao and Jie Wu
Submitted to IEEE Transactions on Mobile Computing
- **Multifunctional PVA/PNIPAM conductive hydrogel sensors enabled human-machine interaction intelligent rehabilitation training**
Yanlong Zhao, Xichong Zhang, Yilin Hao, Yinghe Zhao, Peng Ding, Wei Zhai, Kun Dai, Guoqiang Zheng, Chuntai Liu, Changyu Shen
Advanced Composites and Hybrid Materials (SCI Q1, IF 21.8)
- **Eco-Friendly Multifunctional Hydrogel Sensors Enabled Sustainable and Accurate Human-Machine Interaction System**
Yanlong Zhao, Rui Wu, Yilin Hao, Yi Zhao, Xichong Zhang, Hui Liu, Wei Zhai, Kun Dai, Caofeng Pan, Chuntai Liu, Changyu Shen
Advanced Materials (Top-tier journal in materials science, SCI Q1, IF 26.8)
- **CoSine: Enhancing LLM Serving via Collaborative and Decoupled Speculative Inference**
Luyao Gao, Jianchun Liu, Xichong Zhang, Yunming Liao, Guoju Gao
IEEE International Conference on Computer Communications (INFOCOM), 2026
- **SFedPO: Streaming Federated Learning with a Prediction Oracle under Temporal Shifts**
Jinrui Zhou, Haotian Xu, Xichong Zhang, He Sun, Mingjun Xiao
International Conference on Machine Learning (ICML), 2026
- **SFedKD: Sequential Federated Learning with Discrepancy-Aware Multi-Teacher Knowledge Distillation**
Haotian Xu, Jinrui Zhou, Xichong Zhang, Mingjun Xiao
Under Review

Selected Honors and Awards

- Outstanding Graduate Award, USTC 2025
- Graduate Academic Scholarship, USTC 2022, 2023, 2024
- First Prize in College Physics Experiment IV, USTC 2021
- National Encouragement Scholarship 2019, 2020

Teaching Experience

- Teaching Assistant, 011127, Data Structure, USTC Fall 2022

Skills

- **Languages:** Mandarin (native), English (intermediate)
- **Programming Languages:** Python, C/C++, MATLAB, Bash, Latex
- **Deep Learning Frameworks:** PyTorch, HuggingFace Transformers